How predictable is precipitation from the resolved large-scale state?

A scale-dependent information horizon

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Abstract

Increasing model resolution is widely equated with increasing forecast skill, yet resolving a convective cell does not make it predictable: resolution controls what a model can *represent*, not what the atmosphere will *do*. Parameterised convection makes precipitation a deterministic function of the resolved state, hiding a stochasticity that convection-permitting models expose. We ask how much predictive information the resolved large-scale state carries about precipitation — at what spatial scale, and out to what lead time — a property of the atmosphere that upper-bounds any deterministic forecast system. Using a global convection-permitting simulation (DYAMOND/MPAS, 3.8 km) as truth, we train per-lead neural networks as best-case estimators of the state-to-precipitation map, and evaluate them with an object-based decomposition that separates the precipitation error into *detection* (trigger), *area*, and *intensity* (amplitude). Detection skill collapses monotonically with decreasing system scale, and an exponential fit yields a conditional predictability horizon for the convective *trigger* of only $T_e \approx 0.7$ h at scales below 65 km, rising to $\sim$9 h for systems larger than 200 km. For large systems the *amplitude* remains informative for many hours ($T_e \approx 21$ h), while for small systems amplitude fades as rapidly as the trigger. The only small-scale signal that survives the trigger horizon is the externally forced diurnal cycle: the network stores this forced climatology in its weights and reinjects it even when all solar-tied inputs are removed — substituting climatology for forecast skill once predictability is lost. The same scale-dependent collapse appears in GraphCast, an operational AI model of entirely different architecture and training data, indicating a physical, architecture-independent bound on data-driven precipitation forecasting at convective scales.

1 Introduction

Precipitation predictability spans many orders of magnitude in space and time. A squall line can be warned a day ahead; a single convective cell cannot be warned hours ahead. Both facts are widely accepted, yet the precise boundary between them — at what spatial scale and lead time the resolved large-scale atmospheric state loses its grip on precipitation — remains poorly quantified. The question is not merely academic: flash floods, isolated heavy rainfall, and convective wind damage cluster at the small end of the scale spectrum, where predictability is shortest and warning lead times matter most.

A fundamental obstacle is a widespread modelling artefact. In coarse-resolution numerical weather prediction, convection is parameterised: precipitation becomes a deterministic function of the resolved state, so coarse-model rain inherits the long predictability of the large-scale flow. Global convection-permitting models (CPMs), now routinely run at 3–4 km (Satoh et al., 2019; Stevens, Satoh, Auger, et al., 2019), remove this scheme and let the resolved environment produce many possible realisations of where and when cells fire, recovering the atmosphere's true, scale-dependent predictability limit (Lorenz, 1969; Zhang et al., 2007; Judt, 2018; Hohenegger and Schär, 2007). Increasing horizontal resolution is routinely equated with increasing forecast skill; for precipitation at convective scales, this equation fails, because resolution determines what a model can *represent*, not what the atmosphere will *do*.

Quantifying that limit requires a verification framework that separates distinct failure modes. Standard pixel-wise scores — RMSE, equitable threat score, correlation — apply a double penalty to a displaced precipitating system (it scores as both a miss and a false alarm) and reward spatial smoothing, suppressing the small-scale features that are hardest to predict (Gilleland et al., 2009). Object-based methods instead match precipitating systems and independently score their detection, area, and intensity (Davis, Brown, and Bullock, 2006). Such a decomposition separates the *trigger* failure (wrong location or missed initiation, governed by sub-grid boundary-layer processes that are effectively stochastic at the resolved scale) from the *amplitude* failure (correct location, wrong intensity, governed by large-scale thermodynamics that evolve on the timescale of the flow). These two failures have different physical origins and, as we show, very different predictability timescales.

Data-driven weather models have recently achieved forecast skill competitive with operational NWP for large-scale variables at multi-day leads (Lam et al., 2023), yet precipitation remains a known challenge at convective scales (Ravuri et al., 2021). A neural network trained to minimise mean squared error against a CPM truth predicts the conditional mean of the precipitation distribution — a quantity that is necessarily blurred when the distribution is broad and triggering is stochastic. Understanding whether a given model *cannot* predict small-scale rain, or whether no deterministic model *can*, requires separating architectural limitations from physical ones.

Against this background we address three questions:

1. *How much predictive information does the resolved large-scale state carry about precipitation, and how does this depend on system scale and on the choice of input variables?* We use zero-lead networks as instruments that quantify how much instantaneous environmental information the state carries about rain, and compare predictor sets to isolate the contribution of dynamical versus thermodynamic variables.

2. *How rapidly does this predictive information decay with lead time, and is the decay rate scale-dependent?* We fit an exponential information horizon separately for trigger (detection) and amplitude (intensity) skill across scale bands.

3. *Is the observed scale-dependent collapse specific to our network, or is it a physical, architecture-independent bound?* We apply the same decomposition to GraphCast, a state-of-the-art AI forecast model trained on entirely different data with a different architecture.

Section 2 describes the data, the lead-conditioned network, and the object-based decomposition metric.

Results are presented in Section 3: network performance and variable sensitivity at zero lead (§3.1), the scale-dependent predictability horizon (§3.2), and the generalisation to GraphCast (§3.3). Section 4 discusses physical interpretation, the role of the forced diurnal cycle as a learned climatological prior, and implications for AI precipitation forecasting. Section 5 concludes.

2 Data and Methods

2.1 Data

The dataset is the DYAMOND MPAS-Atmosphere global cloud-resolving simulation, performed at ∼3.8 km horizontal resolution with 15-min output over a 40-day boreal-winter period. MPAS employs a fully compressible, nonhydrostatic dynamical core on an unstructured mesh, enabling explicit simulation of convective processes; cloud microphysics are represented with a double-moment scheme. Global cloud-resolving models explicitly resolve deep convection and provide dynamically consistent four-dimensional fields essential for process-level study and data-driven modelling (Satoh et al., 2019). MPAS generally reproduces mean precipitation and its diurnal cycle within the multi-model DYAMOND range (Stevens, Satoh, Auger, et al., 2019), though it tends to overestimate deep convective clouds and heavy rainfall relative to the IMERG satellite product (Feng et al., 2025). All fields are regridded to a 0.25° longitude–latitude grid and cropped to the tropics (±40°), retaining 15-min snapshots as individual samples.

2.2 Lead-conditioned network

We frame the problem as follows. Let $X(t)$ be the resolved large-scale state at time t and $P_{t+\tau}$ the precipitation field at lead τ. A network that minimises mean squared error over $P_{t+\tau}$ converges to $\mathbb{E}[P_{t+\tau} \mid X(t)]$ — the conditional mean, and the Bayes-optimal deterministic predictor of precipitation given the resolved state. We use this network as an **instrument**: its skill measures how much predictive information the environment carries about the rain; its failure is evidence of genuine physical unpredictability, not model weakness.

Input state. The full state $X(t)$ comprises nine variables spanning the radiative–moist background and the dynamical structure: column water vapour (PW), 2 m temperature (T_{2m}), outgoing longwave radiation (OLR), moist static energy at 500 and 850 hPa (h_{500}, h_{850}), and zonal and meridional winds at 500 and 850 hPa. We compare this full nine-variable set against two reduced five-variable sets — thermodynamics only with moist static energy, and thermodynamics only with relative humidity — to isolate the contribution of the dynamical fields.

Architecture. The network is a 2-D U-Net with skip connections and bilinear upsampling (base width 64 channels, depth 5, dropout 0.05), trained on 64×64 patches tiled from the tropical crop. It has a single softplus output head, enforcing $\widehat{P} \geq 0$.

Training objective. The network minimises mean squared error restricted to rain pixels,

$$\mathcal{L} = \frac{1}{|\Omega|} \sum_{i \in \Omega} (\widehat{P}_i - P_i)^2, \qquad \Omega = \{i : P_i > 0\}, \tag{1}$$

where the sum runs over all raining pixels in the batch. Masking to Ω prevents the heavy rain/no-rain class imbalance from suppressing predicted intensities; the softplus output independently enforces non-negativity. A *separate* network is trained for each lead $\tau \in \{0, 0.25, 0.5, 0.75, 1, 1.5, 2, 3, 4, 6\}$ h, removing autoregressive error accumulation; the skill-versus-lead curve is a clean measurement of state-to-rain predictability alone.

2.3 Object-based precipitation decomposition

We evaluate forecasts at the level of precipitating *systems* rather than grid points. For a single 2-D field at one time we threshold at $r_0 = 0.1\,\mathrm{mm\,h^{-1}}$ and label connected components with 8-connectivity. For each system we accumulate the precipitating area A ($\mathrm{km^2}$), the total precipitation $P = \sum \mathrm{rate} \cdot \mathrm{area}$ ($\mathrm{mm\,km^2}$), and the area-weighted mean intensity $\rho = P/A$ ($\mathrm{mm\,h^{-1}}$), with $L = \sqrt{A}$ as the characteristic scale. Defining ρ as the area-weighted mean makes the identity exact,

$$P = \rho\, A, \qquad \ln\frac{P_p}{P_t} = \underbrace{\ln\frac{A_p}{A_t}}_{\Delta \ln A} + \underbrace{\ln\frac{\rho_p}{\rho_t}}_{\Delta \ln \rho}, \tag{2}$$

so the precipitation error of a matched system splits exactly into an *area* error $\Delta \ln A$ and an *intensity* error $\Delta \ln \rho$, with no cross-term. The log form is scale-free, so a $+50\%$ area error on a 30 km cell and on a 400 km complex contribute comparably.

Matching. Within each frame, predicted systems are matched to truth systems by greedy descending intersection-over-union (IoU), accepting a pair when IoU ≥ 0.1. This yields one-to-one matches and three roles: *match* (truth+pred), *miss* (truth system with no predicted counterpart), and *false alarm* (predicted system with no truth counterpart). A system that is displaced but still overlapping is a *match* carrying area/intensity error, not a miss–false-alarm pair: the matching tolerates sub-system displacement, avoiding the double penalty of pixel-wise scores.

Skill metrics and predictability horizon. Stratified into log-spaced scale bins, we compute per bin the probability of detection $\mathrm{POD} = n_{\mathrm{match}}/(n_{\mathrm{match}} + n_{\mathrm{miss}})$, the false-alarm ratio $\mathrm{FAR} = n_{\mathrm{fa}}/(n_{\mathrm{match}} + n_{\mathrm{fa}})$, and the precipitation-weighted area and intensity biases $\Delta \ln A$, $\Delta \ln \rho$ for matched systems. To summarise the decay of skill $\mathcal{S}(\tau)$ with lead we fit

$$\mathcal{S}(\tau) \approx (\mathcal{S}_0 - \mathcal{S}_\infty)\, e^{-\tau/T_e} + \mathcal{S}_\infty, \tag{3}$$

yielding a zero-lead skill $\mathcal{S}_0$, an e-folding predictability time T_e, and an asymptotic floor $\mathcal{S}_\infty$ set by climatology or persistence.

3 Results

3.1 Network performance at zero lead and variable sensitivity

At zero lead, OLR directly tracks cloud-top temperature, PW reflects column saturation, and T_{2m} captures the boundary-layer thermodynamic state. Zero-lead performance therefore measures how much

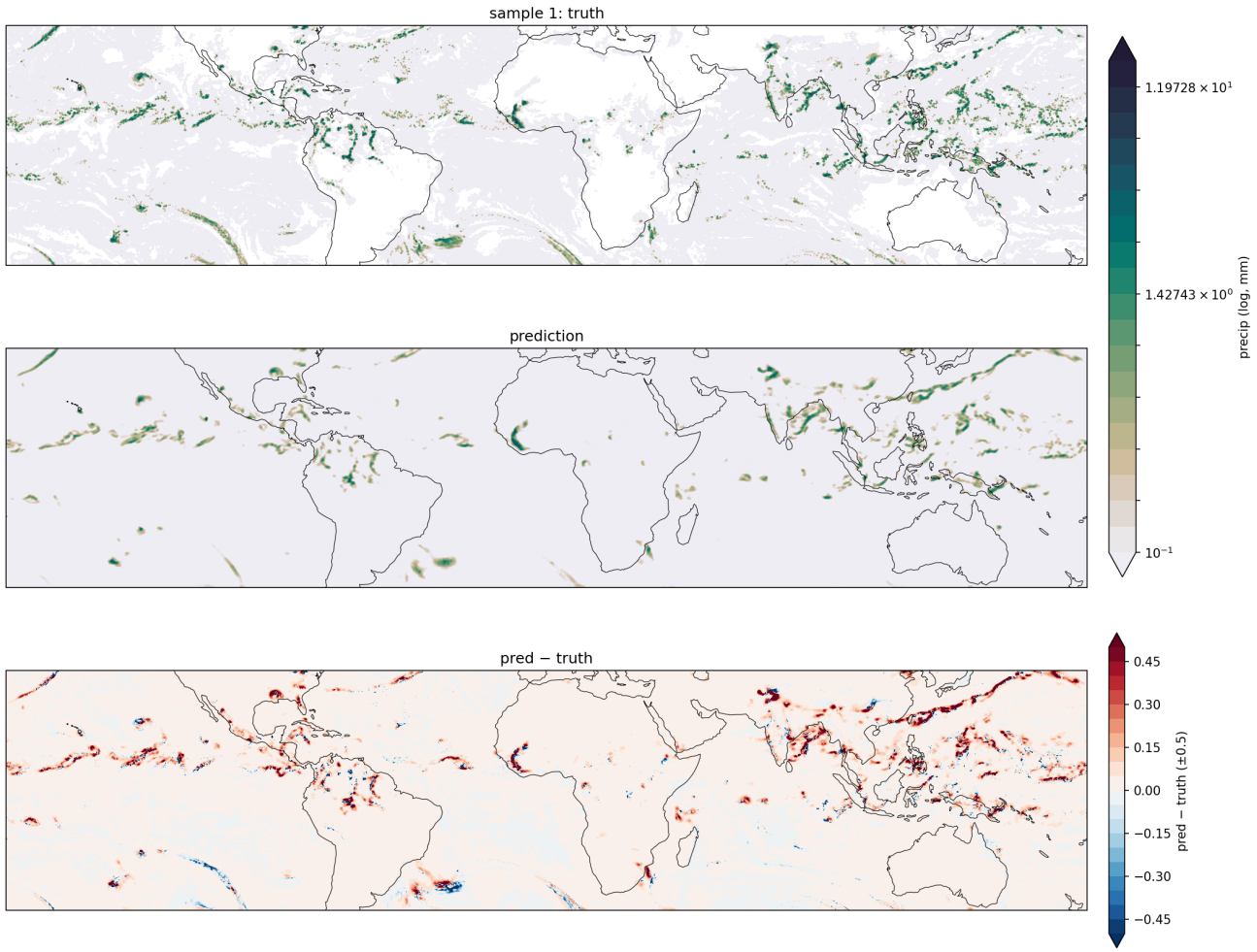


Figure 1: Tropic-wide precipitation at 3 h lead: truth (left), prediction (centre), and difference (right) for one held-out test frame. The prediction captures the large-scale organisation but is smoother than the truth; the difference field is dominated by displaced and under-predicted small-scale features.

predictive information the resolved state carries about the current precipitation field — an upper bound on what any lead-τ network can achieve.

Figure 1 shows the prediction at 3 h lead for context: the network reproduces the large-scale organisation of precipitation — the ITCZ, major convective envelopes — but is visibly smoother than the truth, with the difference field dominated by small-scale structure. This smoothing is expected: when triggering is stochastic, the conditional mean is a spatial average over all possible realisations of where cells could fire.

The object-based decomposition on a single mid-afternoon scene over West Africa (Fig. 2) makes the failure mode concrete: the model captures large organised convective bands but is blind to the scatter of small ($\lesssim 50$ km) cells along the ITCZ, which are missed rather than displaced.

Variable sensitivity. Figure 3 compares the rain-rate contingency at leads 0 and 1 h for three input configurations. At zero lead, the full nine-variable model (MSE + winds) substantially outperforms the five-variable thermodynamics-only configurations: correlation rises to $r = 0.93$ from $r \approx 0.87$

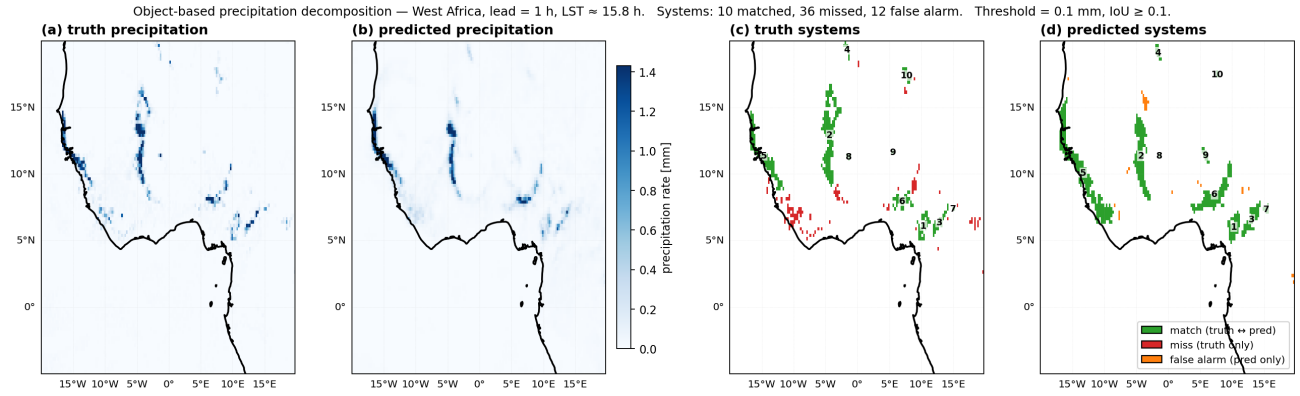


Figure 2: Object-based decomposition on one scene (West Africa, 1 h lead, local solar time ≈ 16 h). (a) truth precipitation; (b) prediction; (c) truth systems coloured green = matched, red = missed; (d) predicted systems coloured green = matched, orange = false alarm. The model captures the large organised bands and misses the small ITCZ cells.

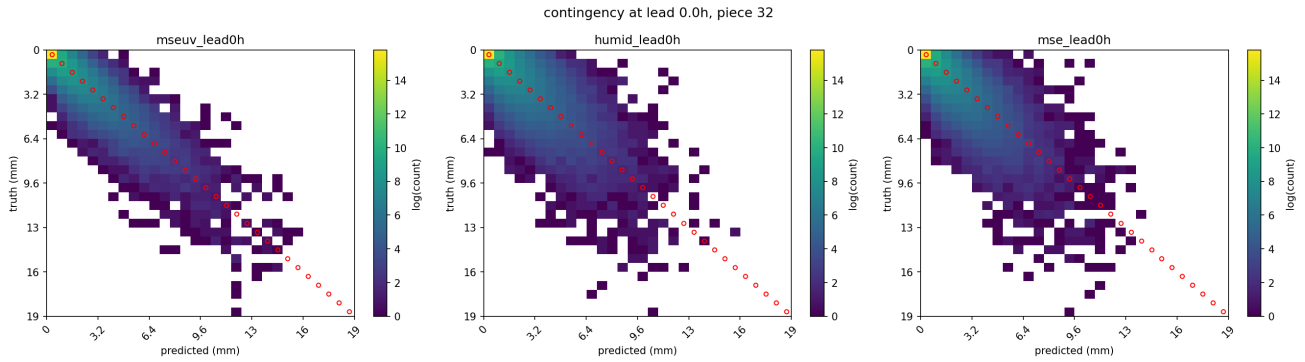
in both reduced sets, and the false-alarm ratio drops from 0.31–0.36 to 0.26. The improvement is attributable to the wind fields: dynamical convergence at 500 and 850 hPa provides direct information about where convection is currently active, information not recoverable from thermodynamic scalars alone. Replacing moist static energy with relative humidity at the same five-variable count produces negligible difference, indicating that the thermodynamic form of the moisture variable matters less than whether the dynamical fields are included. The advantage of the nine-variable set narrows with lead time — winds decouple from future convective location faster than thermodynamic variables — motivating our focus on the nine-variable model as the best-case estimator.

3.2 Scale-dependent predictability horizon

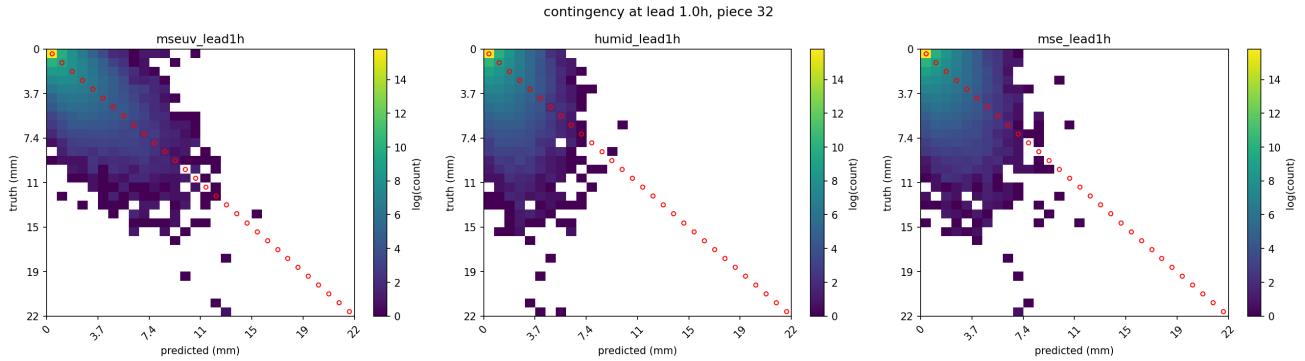
The full scale decomposition at 3 h lead (Fig. 4) reveals how severely detection skill depends on system size. Probability of detection rises monotonically from near zero for the smallest systems ($L < 65$ km) to near unity for the largest ($L > 200$ km); the false-alarm ratio mirrors it. For matched systems, predicted objects are systematically too large ($\Delta \ln A > 0$) and too weak ($\Delta \ln \rho < 0$); at small scales the precipitation budget is dominated by the detection term (missed precipitation), not by distortion of correctly placed systems.

The collapse with lead is exponential and strongly scale-dependent (Fig. 5, Table 1). For the trigger (detection), T_e rises from 0.73 h for small systems ($L < 65$ km) to 8.8 h for large systems ($L > 200$ km) — a factor of twelve. Small convective initiation is forgotten by the resolved state in under an hour; large organised systems remain locatable for nearly a day.

For the amplitude (intensity), the picture is more nuanced. Large systems retain informative intensity prediction far beyond the 6 h window ($T_e \approx 21$ h): once a large system is correctly located, its predicted intensity is constrained by the slowly evolving large-scale thermodynamic environment. Small systems, however, lose amplitude skill at nearly the same rate as they lose trigger skill ($T_e \approx 0.91$ h), because small-cell intensity is sensitive to the same sub-grid boundary-layer fluctuations that govern whether the cell fires at all. The trigger/amplitude distinction in predictability therefore applies primarily to large systems; for convective-scale objects, both trigger and amplitude are effectively unpredictable beyond



(a) lead 0 h



(b) lead 1 h

Figure 3: Rain-rate contingency (log counts) between truth (y) and prediction (x) at lead 0 h (top) and 1 h (bottom) for three input configurations (9var: MSE + winds; 5var-mse: MSE only; 5var-relhum: relative humidity only). Mass below the diagonal indicates intensity under-prediction. Adding winds substantially reduces false alarms and improves intensity estimation at zero lead.

$\sim$1 h.

The aggregate (all-scale) curves in Fig. 5 reflect an important asymmetry in how scales are weighted by each metric. Trigger skill (POD) is count-weighted: small systems vastly outnumber large ones, so the all-scale POD is dominated by the many missed small cells and closely tracks the small-scale curve. Amplitude skill is instead computed only over *matched* systems; because small systems are rarely matched (their POD is near zero), the matched pool is dominated by the large systems that are successfully located, pulling the all-scale intensity curve toward the large-scale result. The aggregate numbers in Table 1 should therefore be read as: trigger skill is effectively the worst case (small-system performance), while aggregate amplitude skill reflects the best case (large-system performance).

The trigger e-folding time $T_e^{\mathrm{det}} \sim 0.7$ h at convective scales is an upper bound set by the atmosphere: it measures how fast the resolved state loses predictive information about where the next cell fires, regardless of forecast architecture.

3.3 Architecture independence: comparison with GraphCast

The behaviour is not specific to our model or training data. Applying the same object-based decomposition to GraphCast (Lam et al., 2023) precipitation at 6 h lead (verified against ERA5, with a

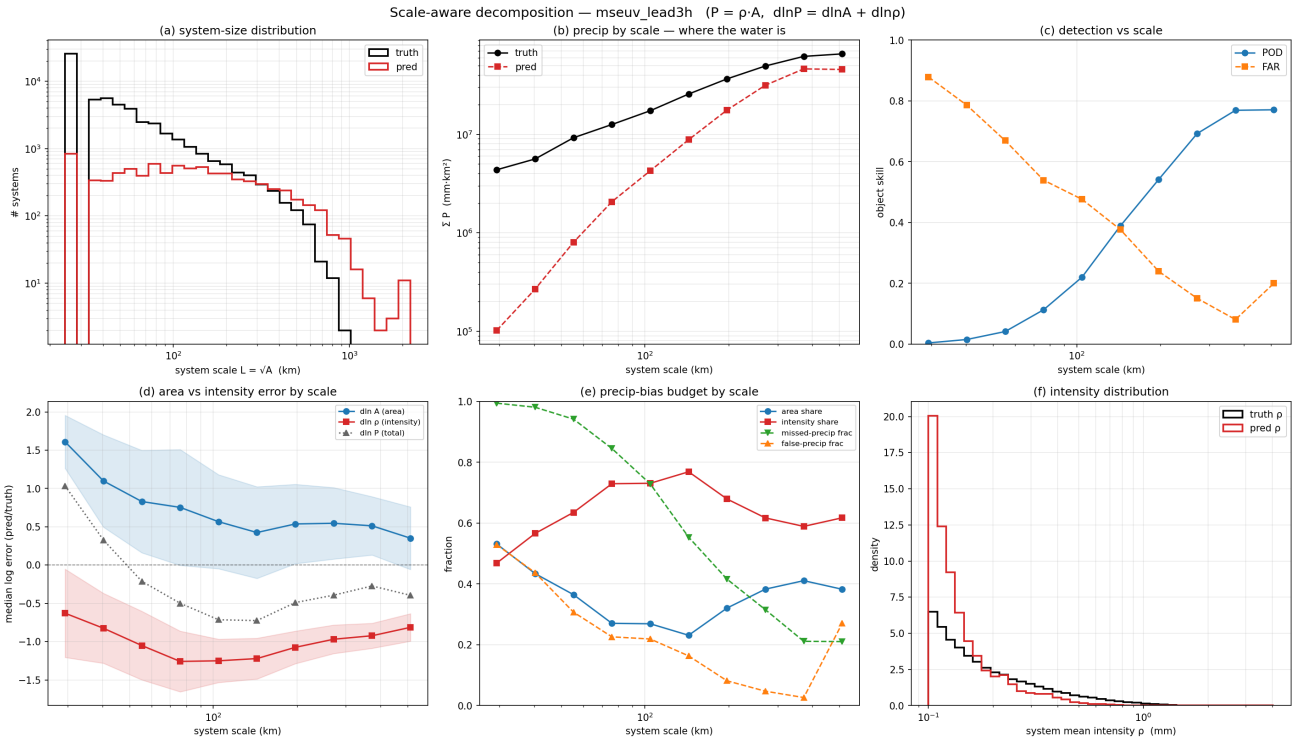


Figure 4: Scale-aware decomposition for the nine-variable model at 3 h lead. (a) system-size distribution, truth vs prediction; (b) precipitation carried by each scale; (c) POD and FAR vs scale; (d) median area ($\Delta \ln A$) and intensity ($\Delta \ln \rho$) bias vs scale; (e) precipitation-bias budget (area vs intensity share, plus missed/spurious fractions); (f) intensity distribution. Detection of small systems is near zero; their error budget is dominated by missed precipitation, not by distortion of correctly located systems.

rate-matched detection threshold of 0.4 mm h^{-1}) reproduces the same signature (Fig. 6): POD rises monotonically with scale and small systems are essentially undetected. GraphCast was trained on ERA5 reanalysis using a graph-neural-network architecture with no shared design choices with our U-Net, verified against a different truth (ERA5 vs MPAS), and evaluated at a different lead and accumulation window (6 h vs 15 min). The robust, comparable quantity is the *shape* of the scale-dependence curve, not its absolute level; both models occupy the same scale-collapse regime. This indicates that "large-system detector" is a property of data-driven precipitation forecasting at convective scales, not of any one network.

4 Discussion

Trigger vs amplitude: two regimes. Our results separate precipitation predictability into two physically distinct components. The *amplitude* — how intense a system will be — inherits the slow memory of the large-scale thermodynamic environment (column water vapour, CAPE, radiative forcing), which evolves on timescales of hours to days. This is why large-system intensity remains informative far beyond the 6 h window ($T_e \approx 21$ h). The *trigger* — whether and where a cell fires — is governed by sub-grid processes (boundary-layer thermals, cold-pool collisions, mesoscale convergence features) that are effectively stochastic at the resolved 0.25° scale. For small convective cells the two regimes collapse: trigger and amplitude skills decay at comparable rates ($T_e \sim 0.7$–0.9 h), because small-cell intensity is

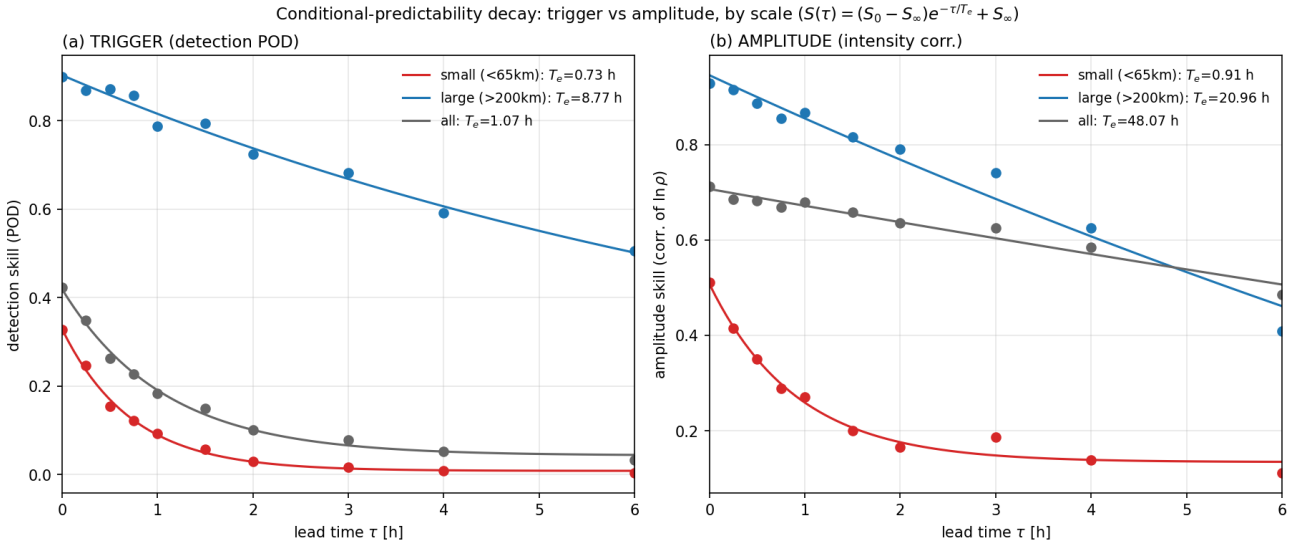


Figure 5: Predictive-information decay vs lead, by scale and component. (a) Detection skill (POD) and (b) intensity skill (correlation of $\ln \rho$ for matched systems), with exponential fits (Eq. 3). The all-scale trigger curve tracks the small-scale curve because POD is count-weighted and small cells dominate the system population; the all-scale amplitude curve tracks the large-scale curve because the matched pool (on which intensity is evaluated) is dominated by the large systems that are successfully detected. Trigger skill for small systems ($T_e^{\text{det}} = 0.73$ h) is a factor of twelve shorter than for large systems ($T_e^{\text{det}} = 8.8$ h). Amplitude of large systems persists well beyond 6 h ($T_e^{\text{int}} \approx 21$ h), while small-system amplitude fades at a similar rate to the trigger ($T_e^{\text{int}} = 0.91$ h).

Table 1: Fitted predictability parameters (Eq. 3): zero-lead skill S_0, e-folding time T_e, and climatological floor S_∞. Large-system intensity T_e exceeds the 6 h window and is poorly constrained (denoted $\gg 6$).

band	term	S_0	T_e [h]	S_∞
small (< 65 km)	detection	0.33	**0.73**	0.01
small (< 65 km)	intensity	0.51	0.91	0.13
large (> 200 km)	detection	0.90	**8.77**	0.09
large (> 200 km)	intensity	0.95	$\gg 6$	—
all	detection	0.42	1.07	0.04
all	intensity	0.71	$\gg 6$	—

sensitive to the same sub-grid fluctuations that determine whether initiation occurs at all.

The forced diurnal cycle: what survives the trigger horizon. Once the lead exceeds the trigger horizon, the only predictable component of small-scale convection is the externally forced diurnal cycle. Figure 7 shows that over land, detection of small systems (POD) is phase-locked to the afternoon convective maximum, while over ocean detection is flat in local solar time — a pattern that persists across leads well beyond T_e. The model is not predicting rain from its inputs at long leads; it is reinjecting a learned climatological prior.

The ablation experiment (Fig. 8) confirms this mechanism. Replacing the solar-tied inputs (T_{2m}, low-level MSE, and OLR) with their per-pixel daily means before inference removes the instantaneous phase signal while preserving geographic structure. The predicted afternoon land peak survives all ablations, showing the diurnal cycle is encoded in the model weights rather than read from the instantaneous

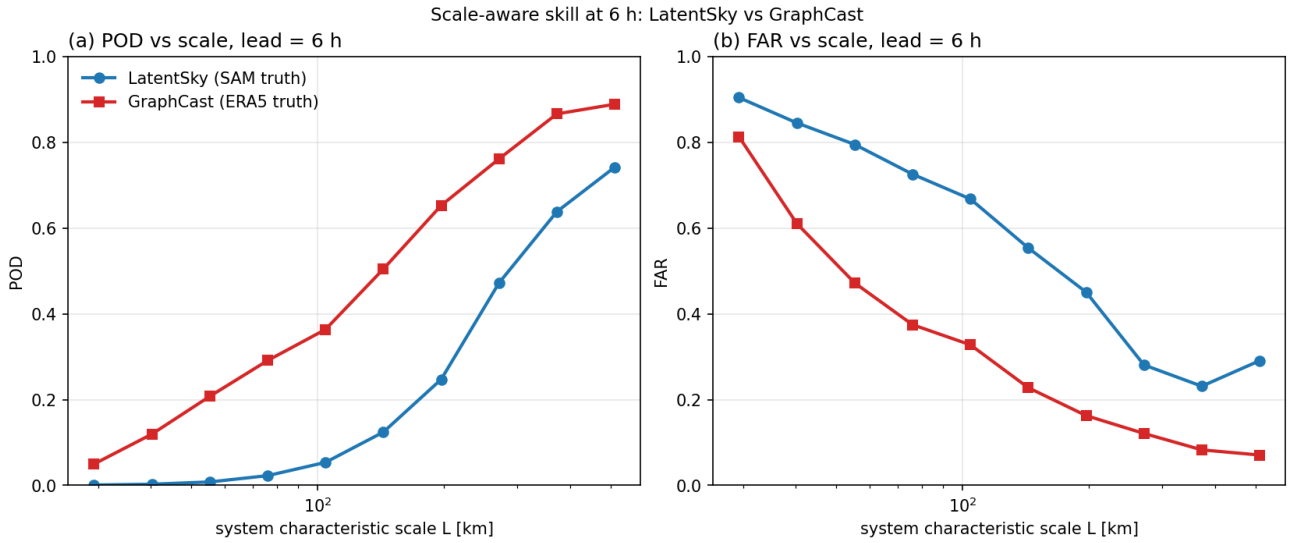


Figure 6: POD (a) and FAR (b) vs system scale at 6 h lead for our model (MPAS truth) and GraphCast (ERA5 truth), at a rate-matched detection threshold (0.4 mm h^{-1}). Both show the monotonic scale-dependent detection collapse. Absolute levels differ because of the different truths and accumulation windows; the robust quantity is the shape of the curve.

inputs.

This has a practical implication: standard pixel scores cannot distinguish climatological reinjection from genuine forecast skill, because both produce temporally correlated errors that inflate the apparent score. The object decomposition can make this distinction: a model that substitutes learned climatology for forecast skill produces detection skill that flattens to the asymptotic floor $\mathcal{S}_\infty$, rather than continuing to decay, and this floor is visible in Table 1. Whether the diurnal residual is read as a positive result (the forced component of convection is correctly recovered) or a cautionary one (reported skill is inflated by climatology) depends on the application; the decomposition makes the distinction transparent in a way that pixel scores cannot.

Implications for AI precipitation forecasting. The convective-scale trigger horizon $T_e \sim 0.7$ h is set by the atmosphere, not by the model: no improvement to the state-evolution core of a deterministic system — better architecture, more training data, higher spatial resolution — can cross it, because the limit lies in the predictive information content of the resolved state itself. That GraphCast shows the same scale-dependent collapse is consistent with this bound being physical rather than architectural. Below the horizon, the conditional distribution of precipitation is genuinely broad; the only routes to better small-scale forecasts are (i) *stochastic* outputs that sample the conditional distribution rather than predict its blurred mean (Ravuri et al., 2021), and (ii) new predictors carrying sub-grid trigger information (dense surface observations, lightning, radar). The smoothing bias we observe — predicted systems too large and too weak — is exactly what a deterministic model minimising MSE against an unpredictable target should produce.

Limitations. The analysis uses a single 40-day boreal-winter MPAS simulation; MPAS overestimates heavy convective rainfall relative to IMERG, and "resolved" is defined by the 0.25° coarsening, so the

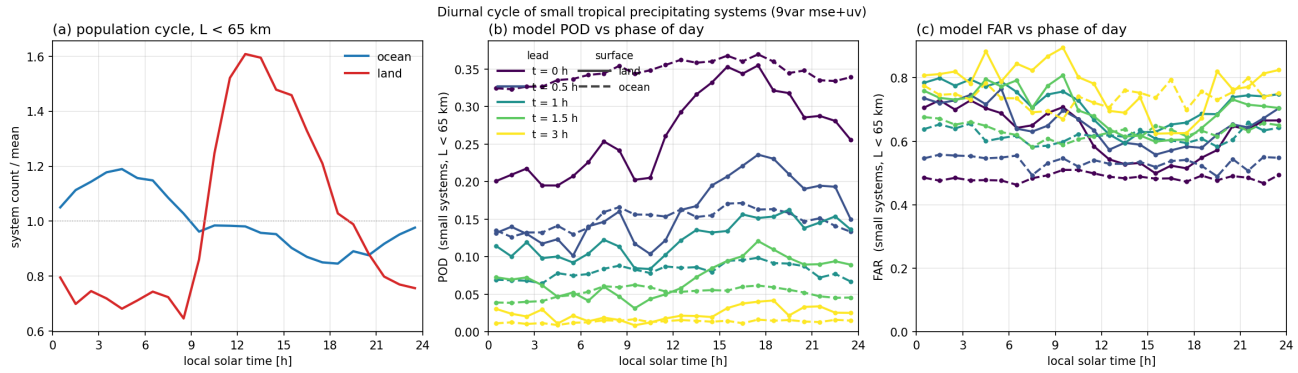


Figure 7: Diurnal cycle of small ($L < 65\,\mathrm{km}$) precipitating systems. (a) truth population cycle vs local solar time, land vs ocean; (b) model POD vs local solar time at several leads, land (solid) vs ocean (dashed). Land detection is phase-locked to the afternoon convective maximum; ocean detection is flat.

trigger horizon is conditional on that scale and on the nine-variable state used here. The GraphCast comparison mixes truth sources (MPAS vs ERA5) and accumulation windows (15 min vs 6 h); the robust statement is the shape of the scale dependence, not absolute skill levels. Natural next steps are an identical-twin intrinsic-predictability comparison, verification against IMERG observations, and a generative model to test whether sampling the conditional distribution recovers the small-scale rain that a deterministic network smooths away.

5 Conclusion

Resolution does not equal predictability. Using a global convection-permitting simulation as truth and per-lead neural networks as best-case state-to-rain estimators, we have measured how much predictive information the resolved large-scale environment carries about precipitation, how fast that information fades with lead, and whether the result is specific to one model.

For the trigger (detection), predictability e-folds in 0.73 h at convective scales ($L < 65\,\mathrm{km}$) and in 8.8 h for large organised systems ($L > 200\,\mathrm{km}$) — a factor of twelve. For the amplitude (intensity), large systems remain predictable far beyond the 6 h window, while small systems lose amplitude skill as rapidly as they lose trigger skill. Past the trigger horizon, the only surviving small-scale signal is the externally forced diurnal cycle, which the network encodes as a learned climatological prior and reinjects regardless of the instantaneous inputs; the object-based decomposition reveals this substitution where pixel scores cannot. The same scale-dependent detection collapse appears in GraphCast, indicating a physical, architecture-independent bound on data-driven precipitation forecasting at convective scales.

For the design of AI precipitation systems, the implication is clear: deterministic minimisation of MSE cannot close the convective gap. Stochastic outputs and new sub-grid predictors are the two routes beyond the predictability horizon set by the resolved state.

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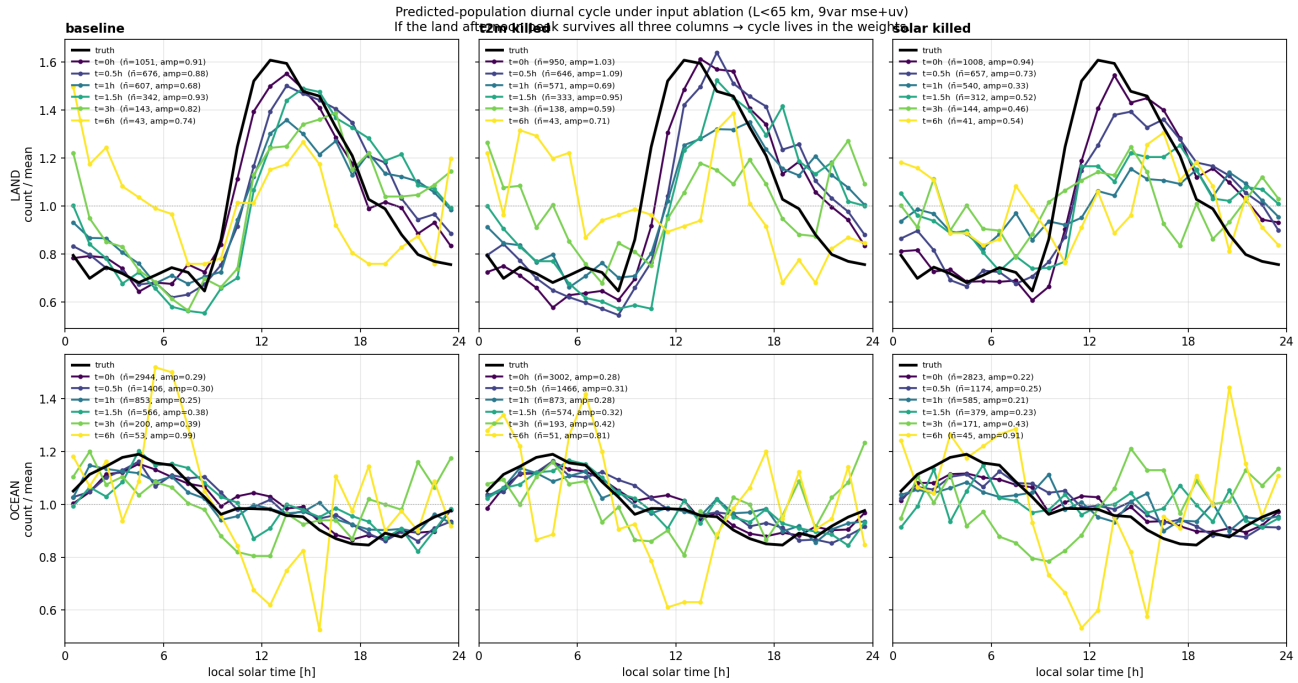


Figure 8: Input-ablation test. Predicted small-system population cycle vs local solar time (top: land, bottom: ocean) for the baseline and two ablations in which solar-tied inputs are replaced by their daily mean. The land afternoon peak survives all ablations, showing the diurnal cycle is encoded in the model weights.

Data availability. The DYAMOND MPAS simulation is available through the DKRZ long-term archive. ERA5 reanalysis is available through the Copernicus Climate Change Service. GraphCast outputs were produced following Lam et al. (2023).

Code availability. Code for the LatentSky network and the object-based decomposition will be released upon acceptance at [repository URL to be provided].

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